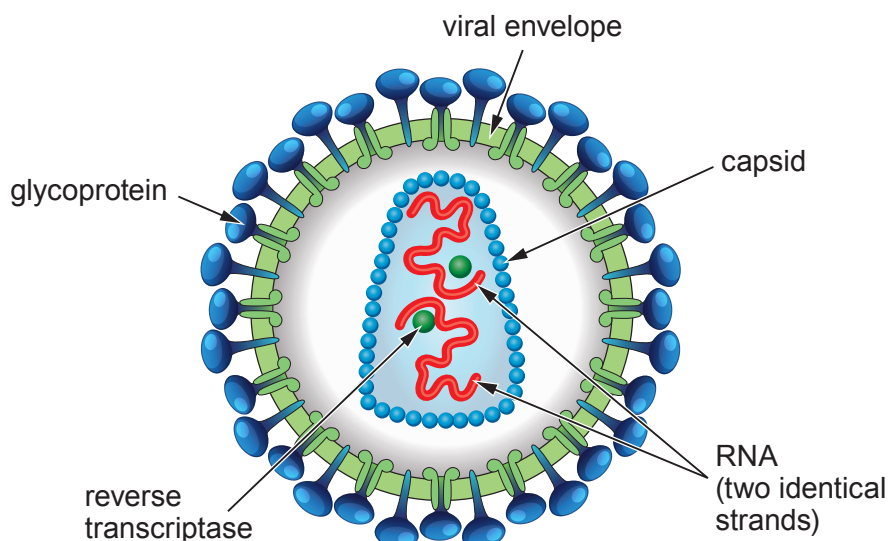


5. The Human Immunodeficiency Virus (HIV) was identified in the 1980s. It targets cells called lymphocytes which are part of the immune system. **Image 5.1** shows a diagram of HIV.

Image 5.1



- HIV contains reverse transcriptase, an enzyme involved in viral RNA replication.
 - RNA and reverse transcriptase are contained within a capsid which is surrounded by a viral envelope.
 - The viral envelope contains a phospholipid bilayer.
 - Glycoproteins, projecting from the viral envelope, allow the virus to infect lymphocytes.
- (a) (i) Use your knowledge of the one gene one polypeptide hypothesis and **Image 5.1** to identify the **minimum** number of genes within the viral RNA. Explain your answer. [3]

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- (ii) Use your knowledge of viral structure to give **two** reasons why HIV cannot reproduce without infecting a host cell. [2]

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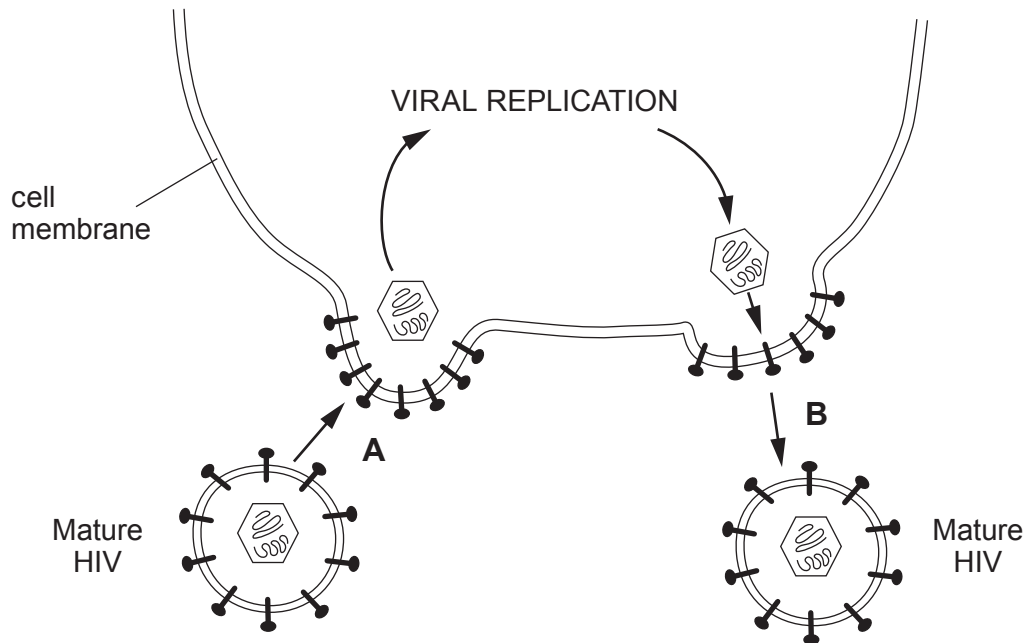
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- (b) **Image 5.2** shows how HIV infects a human cell and makes copies of itself.

Image 5.2



- (i) Assuming the virus is spherical and that the diameter of HIV across the viral envelope is 120 nm, use the formula below to calculate the surface area of HIV. [2]

Surface area of a sphere = $4\pi r^2$.
 $\pi = 3.14$

Surface area = nm²

- (ii) With reference to **Image 5.2**, state the names of the processes occurring at **A** and **B**. Explain the effect each process has on the total surface area of the infected cell. [2]

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